Merugumala Jagadeesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.88
Narayana JR College	Class XII (Telangana State Board)	2024	97.70%
ViswaBharati E.M.H School	Class X (AndhraPradesh SSC Board)	2022	98.00%

SKILLS

Languages: C , C++ , Go , Python , SQL

Frameworks & Libraries: Gin, Cobra, React, Next.js

Databases: PostgreSQL, Redis

DevOps & Tools: Docker, GitHub Actions, Git,Linux, Postman, VS Code

RELEVANT COURSES

Computer Science: Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Architecture, Compilers, Software Development Fundamentals, Discrete Mathematics

Mathematics: Calculus 1 & 2, Linear Algebra, Transform Techniques, Probability and Random Variables

PROJECTS

xv6 OS Kernel extensions Github

- Redesigned xv6's CPU scheduler by replacing Round Robin with a custom 4-level System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ), implementing syscall-aware priority adjustment, starvation prevention, and per-process scheduling statistics.
- Extended xv6's virtual memory subsystem with a global frame table and Clock page replacement algorithm, implementing scheduler-aware page eviction and exposing per-process memory statistics through a custom getvmstats() system call.
- Designed and implemented a disk-backed swap subsystem, replacing in-memory swapping with FCFS and SSTF disk scheduling using a simulated disk-head latency model.
- Implemented RAID 0, RAID 1, and RAID 5 across four simulated disks using striping, mirroring, and distributed XOR parity, integrating fault-tolerant storage with the kernel swap subsystem.

RISC-V 5 stage pipeline simulator Github

- Developed a cycle-accurate RISC-V 5-stage pipeline simulator (IF, ID, EX, MEM, WB), modeling pipeline registers and instruction flow across all execution stages.
- Implemented operand forwarding, stall insertion, and hazard detection logic to correctly execute 50+ RISC-V instructions while resolving data and control hazards.
- Designed instruction-level execution sequencing and pipeline control mechanisms to accurately simulate realistic processor behavior and pipeline state transitions.
- Validated pipeline correctness using comprehensive test programs covering arithmetic, memory, branch, and control-flow instructions.

Authforge (In Progress) Github

- A production-inspired Identity & Access Management service in Go using layered architecture, implementing JWT authentication, refresh-token rotation, email verification, password reset and bcrypt hashing.
- Designed Redis-backed session management and TTL-based token storage for refresh tokens, email verification, password reset, and login rate limiting to protect authentication endpoints from brute-force attacks.
- Containerized the Go API, PostgreSQL, and Redis using Docker Compose and managed database schema evolution with Goose migrations.
- Currently implementing a reusable Role-Based Access Control (RBAC) framework with configurable roles and permissions for seamless integration across backend applications.

Gommit Github

- A cross-platform AI-powered CLI in Go that generates Git commit messages with support of multiple LLM providers, including OpenAI, Gemini, Ollama, and Groq.
- Designed a preprocessing pipeline that filters irrelevant Git diffs before sending code to the LLM, reducing unnecessary context and improving prompt efficiency.
- Automated cross-platform builds and releases for Linux, macOS, and Windows using GitHub Actions, and achieved 200+ GitHub repository clones after open-source release.

Arbitrary Precision Arithmetic Library Github

- Built an arbitrary-precision arithmetic library in Java supporting large integer and decimal computations without using BigInteger or BigDecimal.
- Implemented custom algorithms for arithmetic operations using string-based number representations.
- Designed reusable OOP-based abstractions (AInteger, AFloat) and containerized the project with Docker.

POSITIONS OF RESPONSIBILITY

Teaching Assistant(TA) for course Discrete Mathematics , Dept of Computer Science

Dev Core at Lambda, The software Development Club of IIT Hyderabad

Web Manager, **Entrepreneurship Cell, IIT Hyderabad**

Sunshine Buddy, **Counselling Cell, IIT Hyderabad**

EXTRACURRICULAR

- Secured **2nd Place** in the **CSE Day Cricket Tournament**.
- Secured **2nd Place** in the **Dodgeball Tournament** organized by the **EBSB Club**